

MATH 3310 HOMEWORK ASSIGNMENT 9

DUE ON THURSDAY 3 NOVEMBER 2022

Please turn in your homework in class or to Weeks Hall 347 no later than 5 pm.

- (1) Give a proof of the following statement:

The inequality $2^{n-1} \leq 2^n - 1$ holds for all $n \in \mathbb{N}$.

- (2) Give a proof of the following statement:

The inequality $2n + 1 \leq 2^n$ holds for all integers $n \geq 3$.

- (3) Consider the following induction argument.

In any subset of $\mathbb{Z}$ with only one element, all elements trivially have the same parity. Now, let $k \geq 1$ and assume that for every subset of $\mathbb{Z}$ with cardinality k all numbers in the subset have the same parity. Let B be a subset of $\mathbb{Z}$ with $|B| = k + 1$. Write $B = \{b_1, \dots, b_k, b_{k+1}\}$. The subsets $B' = B - \{b_1\}$ and $B'' = B - \{b_{k+1}\}$ both have cardinality k , so in either set all elements have the same parity. Further, as B' and B'' have elements in common, the elements in B' and B'' have the same parity.

The argument proves (mark all that apply):

- (a) In a finite subset of $\mathbb{Z}$ all elements have the same parity.
 - (b) In a non-empty finite subset of $\mathbb{Z}$ all elements have the same parity.
 - (c) Nothing, the induction base is wrong.
 - (d) Nothing, the induction hypothesis is wrong.
 - (e) Nothing, the induction step is wrong.
- (4) Prove or disprove each of the following statements.
- (a) There exist integers x and y such that $12x + 9y = 100$ holds.
 - (b) Let $x, y \in \mathbb{R}$. If $x < y$ then $x^2 < y^2$.
 - (c) Let $n \in \mathbb{Z}$. If n is prime, then $2^n - 1$ is prime.
 - (d) Let $a, b \in \mathbb{Z}$. If a and b are even, then $4 \mid (a^2 + b^2)$.

- (5) Prove the following statement:

There are no integers x and y with $x^2 = 4y + 2$.